MEMORANDUM

6/3/2026

TO: Joseph Pinheiro, Professor, Mechanical Engineering

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SUBJECT: **Lab 7 – Cylinder in a Uniform Flow Stream**

Abstract
This experiment studied flow over a circular brass cylinder in the Cal Poly ME wind tunnel. The goal was to examine the pressure distribution around the cylinder, the velocity profile in the wake, and the drag coefficient. The brass cylinder had a diameter of $D = 1.50$ in and was tested at a Reynolds number of $Re_D = 3.49 \times 10^4$. Surface pressure measurements were taken from 0°to 180°and converted into pressure coefficients. The highest-pressure coefficient was $C_P = 1.03$at the stagnation point, and the lowest value was $C_P = -1.54$near 70°, showing that the flow accelerated around the cylinder and separated near the back side. Wake velocity profiles were measured at $x = 17.5$in and $x = 25.0$in downstream of the cylinder. Both wake profiles showed a velocity drop near the centerline and recovery away from the wake region. The surface pressure method gave a drag coefficient of $C_D = 1.256$, which agreed well with the expected value of about $C_D = 1.2$. The wake momentum method gave much smaller drag coefficients of $C_D = 0.179$and $C_D = 0.031$, which were much lower than expected. These low values were likely caused by inconsistent wake measurements, unsteady flow behind the cylinder, possible probe alignment or pressure zeroing errors, and not capturing the full wake region. Overall, the surface pressure method gave the more reliable drag result, while the wake momentum method underestimated the drag for this experiment.

Background/Theory
When air flows toward a circular cylinder, it first hits the front of the cylinder and slows down almost to zero. At this location, called the stagnation point, the pressure is at its highest. As the air moves around the curved surface, it speeds up and the pressure decreases. Near the back of the cylinder, the flow can no longer stay attached to the surface, so it separates and forms a wake. This wake has lower pressure and lower velocity than the incoming flow, which creates most of the drag on the cylinder.

The Reynolds number is used to describe the type of flow around the cylinder. It depends on the free-stream velocity, the cylinder diameter, and the kinematic viscosity of air. A higher number of Reynolds means the flow has more momentum compared to the effects of viscosity. In this lab, the Reynolds number is calculated to help predict what the flow and wake should look like. For the expected range in

this experiment, the cylinder should produce a wide, turbulent wake and a relatively large drag coefficient. See figure 1.

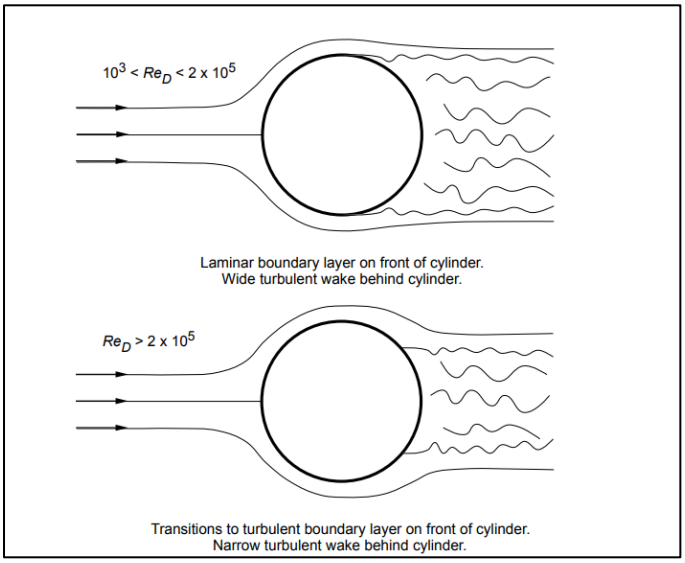


Fig 1. Streamlines for uniform flow over a cylinder for two ranges of Reynolds number, Re_D.

The pressure coefficient C_P, is used to compare the pressure at different points around the cylinder. It shows how much the local pressure changes compared to the free-stream flow. The pressure coefficient should be highest at the front of the cylinder; velocity of the air is zero. It should decrease as the air speeds up around the sides of the cylinder. On the back side, the pressure should stay low because the flow has separated and a wake has formed.

$$C_P = \frac{p - p_o}{\frac{1}{2}\rho U^2} \tag{1}$$

The drag coefficient is used to compare drag in a way that does not depend only on the cylinder size or wind tunnel speed. In this lab, drag is calculated in two ways. The first method uses pressure measurements around the surface of the cylinder. Since the front of the cylinder has higher pressure than the back, this pressure difference creates a force in the flow direction. By adding up the pressure forces around the cylinder, the drag coefficient can be estimated.

The second method uses velocity measurements in the wake behind the cylinder. The cylinder slows down the air behind it, so the wake has a velocity deficit compared to the incoming flow. A larger velocity deficit means more momentum was lost, which means the cylinder caused more drag. The wake velocities are found from the Pitot-static pressure readings, then compared to the free-stream velocity.

The wake is measured at two locations downstream of the cylinder to see how it changes with distance. Closer to the cylinder, the wake is expected to have a larger velocity drop and more uneven flow. Farther downstream, the wake should spread out, and the velocity should start to recover.

Apparatus and Experimental Procedure

The experiment was conducted using the Cal Poly ME wind tunnel facility and in accordance with the lab manual. A circular brass cylinder was mounted horizontally in the test section with its axis perpendicular to the incoming air flow. The cylinder had a pressure tap that allowed the pressure at different angular locations around the surface to be measured. A Pitot-static probe was also used to measure the wind tunnel dynamic pressure and the velocity profile in the wake behind the cylinder. The Pitot-static probe was mounted on an x-y traverse, which allowed it to be moved to specific downstream and vertical positions. Pressure readings were taken using the pressure transducer connected to the wind tunnel instrumentation system.

Before collecting data, the ambient temperature, atmospheric pressure, and zero pressure reading were recorded. These values were later used to calculate air density and correct the pressure measurements. The wind tunnel was then turned on and allowed to reach steady operation. The free-stream dynamic pressure was measured using the Pitot-static probe so that the free-stream velocity and Reynolds number could be calculated.

The first part of the experiment measured the surface pressure around the brass cylinder. The cylinder was rotated until the highest-pressure reading was found, which corresponded to the stagnation point. This location was marked with a piece of tape and was treated as the 0-degree reference position for all subsequent measurements. The cylinder was then rotated in 10-degree increments from 0 to 180 degrees, and the pressure difference was recorded at each angle. These measurements were used to calculate the pressure coefficient around the cylinder and estimate the drag using the surface pressure method.

The second part of the experiment measured the wake velocity profile behind the cylinder. The Pitot-static probe was positioned downstream of the cylinder and moved vertically through the wake. At each vertical position, the pressure reading was recorded and later converted into a local velocity. This process was repeated at two downstream locations so that the wake profile could be compared closer to and farther behind the cylinder. The measured velocities were normalized by the free-stream velocity, and the vertical positions were normalized by the cylinder diameter.

The lab manual addendum instructed that the wake profile measurements should also be repeated using a smaller aluminum cylinder. However, this portion of the experiment was not completed during the lab period. As a result, the data and analysis in this report only include the brass cylinder measurements.

Analysis

The free stream velocity was determined from the measured Pitot-Static tube pressure using Bernoulli's equation.

$$U = \sqrt{\frac{2p_{\text{dynamic}}}{\rho}} \tag{2}$$

Where $p_{dynamic}$ is the free stream dynamic pressure and ρ is the density of air.

The Reynolds number based on the cylinder's diameter was calculated using the following equation:

$$Re_D = \frac{UD}{\nu} \tag{3}$$

Where D is the cylinder diameter and ν is the kinematic viscosity of the working fluid, air.

<u>Pressure Coefficient</u>

The pressure coefficient around the cylinder was calculated by using the pressure coefficient equation.

$$C_p = \frac{p - p_{free}}{\frac{1}{2}\rho U^2} \tag{4}$$

Since the free stream dynamic pressure was taken directly the coefficient of pressure equation can be simplified to the following.

$$C_p = \frac{\Delta p_{cyl}}{q_{free}} \tag{5}$$

where Δp_{cyl} is the corrected cylinder pressure measurement.

Surface Pressure Drag Coefficient

The drag coefficient due to the pressure distribution is usually determined by integrating the pressure coefficient around the cylinder surface using the following equation:

$$C_{D_p} = \int_0^{\pi} C_p \cos\theta \; d \tag{6}$$

where θ is the angular position measured from the stagnation point.
Since our pressure measurements were at set angular locations, the integral was evaluated numerically using the trapezoid rule.

$$\int_a^b f(x)dx \approx \sum_i \frac{f_i + f_{i+1}}{2}(x_{i+1} - x_i) \tag{7}$$

where

$$f(\theta) = C_p \cos\theta \tag{8}$$

Wake Velocity Profile

The velocity at each wake location, x=17.5 in, 25 in, was determined from the measured pitot static pressure. Using Bernoulli's equation:

$$u = \sqrt{\frac{2p}{\rho}} \tag{9}$$

where p is the local dynamic pressure measured within the wake

In order to compare wake profiles at different locations, the velocity was normalized by the free stream velocity

$$\frac{u}{U} \tag{10}$$

The vertical position as normalized using the cylinder diameter

$$\frac{y}{D} \tag{11}$$

where $y = 0$ corresponds to the cylinder's centerline.

Momentum Deficit Drag Coefficient

The drag coefficient was also determined using the wake momentum deficit. By applying the conservation of momentum to the wake, it yields

$$C_D = 2/D \int_0^H (1 - u/U)(u/U)dy \tag{12}$$

Where H is the height of the wake region where data was collected?
The integral was also evaluated numerically using the trapezoid rule

$$\int_a^b f(y)\ dy \approx \sum_{i=1}^{n-1} \frac{f_i + f_{i+1}}{2}(y_{i+1} - y_i) \tag{13}$$

where

$$f(y) = \left(1 - \frac{u}{U}\right)\left(\frac{u}{U}\right) \tag{14}$$

The drag coefficients obtained from the surface pressure method and momentum deficit method were then compared to the expected drag coefficient for a circular cylinder at the calculated Reynolds number.

Results

Table 1. Cylinder Surface Pressure Data

Cylinder Angle, θ	Differential Pressure Measurements, $p - p_0 (in\ H_2O)$	Pressure Coefficient, C_P
0	0.46	1.03
10	0.40	0.89
20	0.26	0.58
30	0.03	0.07
40	-0.21	-0.47
50	-0.45	-1.01
60	-0.62	-1.39
70	-0.69	-1.54
80	-0.59	-1.32
90	-0.55	-1.23
100	-0.54	-1.21
110	-0.55	-1.23
120	-0.56	-1.25
130	-0.58	-1.30
140	-0.59	-1.32
150	-0.61	-1.36
160	-0.61	-1.36
170	-0.63	-1.41
180	-0.64	-1.43

Table one shows that the pressure coefficient was highest at 0°, where $C_P = 1.03$. This makes sense because this point corresponds to the front outermost part of the cylinder, also known as the stagnation point. As the angle increased, C_P decreased, reaching a minimum of -1.54 at around 70°. The negative pressure coefficients along the back side of the cylinder correspond to a low-pressure wake that forms due to flow separation.

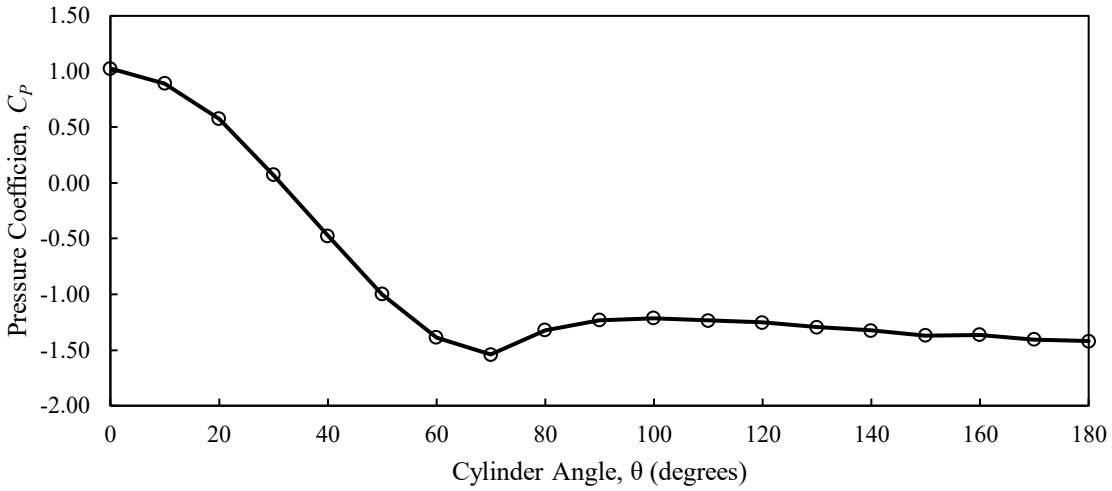


Figure 2. Pressure Coefficient as a Function of Cylinder Angle

Figure 2 shows that the pressure coefficient has highest at 0°, the stagnation point on the cylinder. As the angle increased, we saw a decrease in C_P, with a low point at about 70° followed by a section of similar $C_P$values. The trend for the pressure coefficient makes sense because C_P is highest at the front, then decreases as the flow accelerates and separates from the cylinder, creating a low-pressure zone along the back side. Based on the measurements. The wake likely appears as a broad low-pressure region on the back side of the cylinder. The negative C_P values on the back side show that the flow separated from the cylinder's surface and created a wake. This agrees with the calculated $Re = 3.49\ x10^4$, where we expected a wide region of turbulent wake.

Table 2. Wake Velocity Profile Data (x = 17.5 in)

Streamwise Position, x (in)	Vertical Position, y (in)	Normalized Vertical Position, y/D	Pitot-static Tube Pressure, $(in\ H_2O)$	Local Velocity, $u\ (m/s)$	Normalized Local Velocity, u/U
17.5	16.0	2.67	0.52	14.83	1.079
	15.5	2.33	0.52	14.81	1.077
	15.0	2.00	0.52	14.83	1.079
	14.5	1.67	0.52	14.75	1.073
	14.0	1.33	0.51	14.62	1.064
	13.5	1.00	0.47	14.11	1.026
	13.0	0.67	0.46	13.96	1.016
	12.5	0.33	0.41	13.07	0.951
	12.0	0.00	0.34	12.02	0.875
	11.5	-0.33	0.33	11.72	0.852
	11.0	-0.67	0.38	12.60	0.917
	10.5	-1.00	0.43	13.41	0.976
	10.0	-1.33	0.45	13.82	1.006
	9.5	-1.67	0.48	14.29	1.039
	9.0	-2.00	0.51	14.68	1.068
	8.5	-2.33	0.52	14.80	1.077
	8.0	-2.67	0.52	14.84	1.079

Table 3. Wake Velocity Profile Data (x = 25.0 in)

Streamwise Position, x (in)	Vertical Position, y (in)	Normalized Vertical Position, y/D	Pitot-static Tube Pressure, $(in\ H_2O)$	Local Velocity, $u\ (m/s)$	Normalized Local Velocity, u/U
25.0	16.0	2.67	0.53	14.97	1.089
	15.5	2.33	0.52	14.81	1.078
	15.0	2.00	0.50	14.57	1.060
	14.5	1.67	0.49	14.38	1.046
	14.0	1.33	0.47	14.10	1.025
	13.5	1.00	0.45	13.76	1.001
	13.0	0.67	0.42	13.28	0.969
	12.5	0.33	0.39	12.81	0.932
	12.0	0.00	0.36	12.25	0.891
	11.5	-0.33	0.33	11.80	0.858
	11.0	-0.67	0.38	12.61	0.918
	10.5	-1.00	0.40	13.03	0.948
	10.0	-1.33	0.45	13.79	1.003
	9.5	-1.67	0.48	14.17	1.031
	9.0	-2.00	0.50	14.46	1.052
	8.5	-2.33	0.51	14.70	1.069
	8.0	-2.67	0.52	14.81	1.078

Tables 2 and 3 show the wake velocity profile data at two downstream locations. Both locations had a low in the normalized velocity near *y/D* = 0, the center line of the cylinder. This shows that the cylinder

slowed the flow the most directly behind it, we also noticed that as vertical position moved away from the center our *u/U* crept closer to 1, indicating that the flow was returning to free stream velocity.

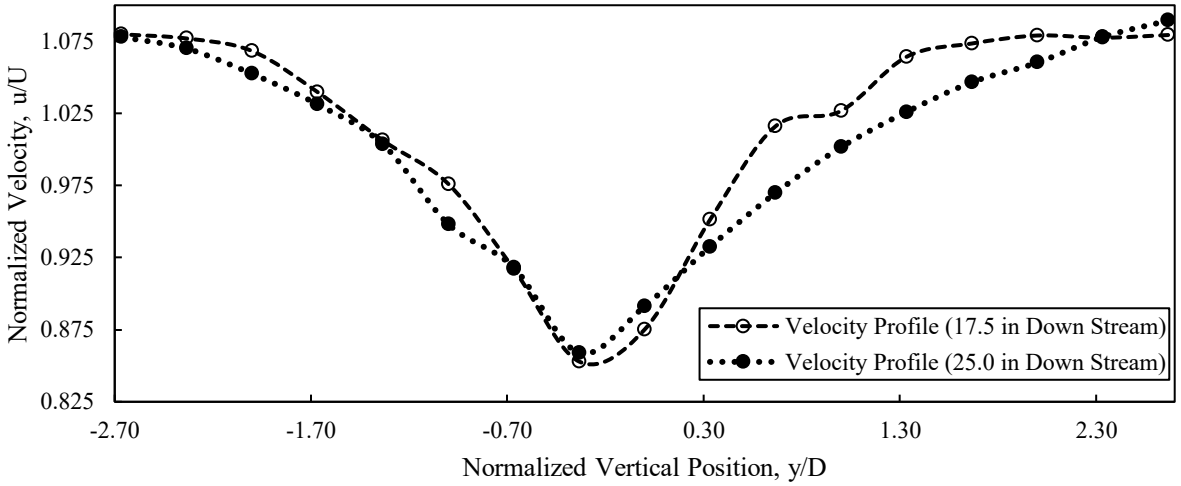


Figure 3. Wake Velocity at Two Downstream Locations

Figure 3 compares the wake profiles at x = 17.5 in and x = 25.0 in downstream. The trends make sense because both profiles show a low point in the normalized velocity in the wake region and a recovery as we move away from the centerline. It is important to note that the farther downstream profile appears broader, and smoother compared to the close profile, this shows that the wake spreads out and begins to mix back into the surrounding flow as it moves downstream.

Table 4. Comparison of Drag Coefficients

Method	Location / Source	Drag Coefficient, C_D
Surface Pressure	Cylinder Surface	1.256
Wake Momentum	x = 17.5 in	0.179
Wake Momentum	x = 25.0 in	0.031
Reference Plot	$Re = 3.49\ x10^4$	≈1.2

Table 4 compares the drag coefficients found from the surface pressure method, the wake momentum method, and the reference plot. For our Reynolds number, the expected drag coefficient was about $C_D = 1.2$. The surface pressure method gave $C_D = 1.256$, which was very close to the expected value. However, the wake momentum method gave much smaller values of $C_D = 0.179$ at $x = 17.5$ in and $C_D = 0.031$ at $x = 25.0$ in. These values are much lower than expected, so the wake momentum method did not give accurate drag results for our data. This likely happened because the wake measurements were taken fairly close to the cylinder, where the flow was still unsteady and uneven. Other possible causes include the Pitot-static probe not being perfectly aligned, a small pressure zeroing error, not measuring across the full wake height, or the equipment being bumped during testing. Some of the measured u/U values were also greater than 1 near the outside of the wake, which suggests that the pressure readings or free-stream reference velocity may not have been completely consistent. Because of these issues, the wake momentum method likely gave drag coefficients that were too small.

Based on the flow visualization, the flow appeared to be attached near the front of the cylinder and the separated near the rear of the cylinder. This separation formed a broad and unsteady wake behind the cylinder. This agrees with the pressure coefficient and wake velocity profiles because we concluded that the back of the cylinder has an area of low pressure. It is also safe to say that as the wake moved farther downstream it becomes more broad and smooth as it mixes back into the flow.

Conclusion

This experiment studied how fluid flows and drags on a circular cylinder in a steady wind tunnel flow. The surface pressure measurements displayed the expected pattern. The highest-pressure coefficient was near the stagnation point at the front of the cylinder. Lower pressure coefficients appeared along the rear, where flow separation and wake formation occurred. The wake velocity profiles also showed expected results, with the lowest normalized velocity found near the centerline of the cylinder. The velocity improved as the probe moved farther away from the wake region. Between the two downstream wake locations, the farther profile appeared broader and smoother. This indicated that the wake spread out and began mixing back into the surrounding flow. The drag coefficient calculated using the surface pressure method was 1.256, which matched well with the reference value of about 1.2. This suggests that the surface pressure method provided a reasonable estimate of the cylinder drag for the tested Reynolds number range. However, the wake momentum method gave much lower drag coefficients of 0.179 at (x = 17.5) in and 0.031 at (x = 25.0) in. These values did not agree with the pressure method or reference value. The difference likely came from the limitations in wake measurements. These included unsteady vortex shedding, the probe being in a highly turbulent wake area, potential pressure zero offset, a limited vertical integration range, and disturbance or misalignment of the instruments during testing. Overall, the experiment demonstrated the relationship between pressure distribution, wake velocity deficit, and drag for flow over a cylinder. The results showed that most of the drag came from the pressure difference between the front and back of the cylinder, which matched with separated flow and turbulent wake formation. The aluminum cylinder portion required by the addendum was not finished during the lab period, so a final comparison between cylinder sizes could not be made. Future tests should include the smaller aluminum cylinder and repeat the wake measurements more carefully. They should also extend the wake survey farther downstream to improve the accuracy of the momentum-based drag calculation.

References

[1] ME 347 Lab Manual

Attachments

1.Raw Data
2. Hand Calculations

Atttachment 1. Raw Data

Cyldiner Pressure Readings

theta (deg)	p-p0 (raw) in/h20
0	0.43
10	0.37
20	0.23
30	0.01
40	-0.24
50	-0.47
60	-0.65
70	-0.72
80	-0.62
90	-0.58
100	-0.57
110	-0.58
120	-0.59
130	-0.61
140	-0.62
150	-0.64
160	-0.64
170	-0.66
180	-0.66

Wave Velocity x=17.5 in

x (in)	y (in)	Pitot-Static(raw)
17.5	16.0	0.49
17.5	15.5	0.49
17.5	15.0	0.49
17.5	14.5	0.48
17.5	14.0	0.47
17.5	13.5	0.44
17.5	13.0	0.43
17.5	12.5	0.37
17.5	12.0	0.31
17.5	11.5	0.29
17.5	11.0	0.34
17.5	10.5	0.39
17.5	10.0	0.42
17.5	9.5	0.45
17.5	9.0	0.48
17.5	8.5	0.49
17.5	8.0	0.49

Wave Velocity x=15 in

x (in)	y (in)	Pitot-Static (in/h20)
25.0	16.0	0.50
25.0	15.5	0.49
25.0	15.0	0.47
25.0	14.5	0.46
25.0	14.0	0.44
25.0	13.5	0.42
25.0	13.0	0.39
25.0	12.5	0.36
25.0	12.0	0.32
25.0	11.5	0.30
25.0	11.0	0.34
25.0	10.5	0.37
25.0	10.0	0.42
25.0	9.5	0.44
25.0	9.0	0.46
25.0	8.5	0.48
25.0	8.0	0.49

Attachment 2. Hand Calculations

Hand Calcs

Free stream velocity

$P_{stat} = 0.4775$ in H_2O

$P_{free} = (0.4775 \text{ in } H_2O)(249.1) = 111.4677 Pa$

$U = \sqrt{\frac{2P_{free}}{\rho}} = \frac{2(111.4677)}{1.180} = 13.74 \frac{m}{s}$

Reynolds number

$D = 0.0341$ in

$U = 13.74 \frac{m}{s}$

$\delta = 1.5 \times 10^{-5} \; m^2/s$

$Re = \frac{UD}{\delta} = \frac{(13.74)(0.0341)}{1.5 \times 10^{-5}}$

$Re = 3.49 \times 10^4$

CAL POLY

Pressure Coefficient

$\theta = 40^\circ$

$P_{P3} = -0.21$ in/H2O

$P_{dynamic} = 0.447$ in/H2O

$C_p = \frac{-0.21}{0.447} = -0.47$

Pressure Drag Integrand

$C_p \cos\theta = -0.47 \cos(40)$

$= -0.363$